THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A capsule is a cylinder of radius 7 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 2112 cm² **(b)** 748 cm² **(c)** 1364 cm² **(d)** 1056 cm²

2. The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 21, 3, x, 10, y, 14, 9 respectively, and total frequency 100, is 35. The value of y is: 1

(a) 24 **(b)** 22 **(c)** 23 **(d)** 20

3. A red die and a blue die are rolled together. The probability that both numbers are even is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{5}{18}$ **(c)** $\frac{2}{9}$ **(d)** $\frac{3}{4}$

4. A toy is a cone of base radius 42 cm and height 56 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 18480 cm² **(b)** 20328 cm² **(c)** 25872 cm² **(d)** 9240 cm²

5. If $\sum f_i = 40$ and $\sum f_i x_i = 1200$, the mean is: 1

(a) 30 **(b)** 300 **(c)** 1160 **(d)** 40

6. A letter is chosen at random from the word 'PERMUTATION'. The probability that it is a vowel is: 1

(a) $\frac{5}{11}$ **(b)** $\frac{6}{11}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{5}{12}$

7. A solid metal cuboid 24 cm by 12 cm by 18 cm is melted and recast into cubes of edge 3 cm. The number of cubes obtained is: 1

(a) 191 **(b)** 576 **(c)** 384 **(d)** 192

8. For the data with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 with frequencies 3, 6, 13, 15, 7, 6 respectively, The mean of the data is: 1

(a) 111 **(b)** 113 **(c)** 115 **(d)** 112

9. A red die and a blue die are rolled together. The probability that the first number is greater than the second is: 1

(a) $\frac{4}{9}$ **(b)** $\frac{7}{18}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{7}{12}$

10. A tent is a cylinder of radius 3.5 m and height 10 m surmounted by a cone of the same radius and height 12 m. The area of canvas (in m^2) needed to make the tent is: 1

(a) 352 m² **(b)** 357.5 m² **(c)** 220 m² **(d)** 137.5 m²

11. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 12, 9, 5, 14, 5, 5 respectively, The median of the data is: 1

(a) 38 **(b)** 40 **(c)** 36.2 **(d)** 30

12. A coin is tossed three times. The probability of getting a head on the first toss is: 1

(a) $\frac{5}{8}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{4}{9}$ **(d)** $\frac{3}{8}$

13. A hemispherical bowl has outer radius 8 cm and inner radius 2 cm. The volume of material used to make the bowl is: 1

(a) 2112 cm³ **(b)** 1056 cm³ **(c)** 528 cm³ **(d)** 3168 cm³

14. For the data with classes 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 with frequencies 5, 13, 5, 2, 4, 11 respectively, The mode of the data is: 1

(a) 85 **(b)** 100 **(c)** 90 **(d)** 80

15. 40 cards numbered 1 to 40 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a composite number is: 1

(a) $\frac{7}{10}$ **(b)** $\frac{27}{40}$ **(c)** $\frac{13}{20}$ **(d)** $\frac{13}{40}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. 36 cards numbered 1 to 36 are placed in a box, mixed well and one is drawn at random. Find the probability that the number drawn is divisible by both 2 and 3. 2

17. A solid is a cube of edge 35 cm surmounted by a hemisphere of diameter 35 cm (the hemisphere rests on a face of the cube). Find the total surface area of the solid. 2

18. The marks obtained by students in a test are given below. 2

Marks	60–70	70–80	80–90	90–100	100–110	110–120	120–130
Number of students	2	16	10	2	4	5	11

Find the mode.

19. A bag contains 6 red, 4 blue, 5 green balls. One ball is drawn at random. Find the probability that it is not green. 2

20. A solid metallic cone of base radius 3 cm and height 96 cm is melted and recast into a solid sphere. Find the radius of the sphere. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The marks obtained by students in a test are given below. 3

Marks	40–50	50–60	60–70	70–80	80–90	90–100
Number of students	4	12	10	19	3	2

Find the mean and the mode (in this order).

22. A red die and a blue die are rolled together. Find the probability that the sum of the numbers is 8. 3

23. A capsule is a cylinder of radius 10.5 cm and length 18 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

24. The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	10–20	20–30	30–40	40–50	50–60	60–70
Number of workers	12	9	5	14	5	5

Find the mean.

OR

A bag contains 8 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. Find the value of x. 3

25. A toy is a cone of base radius 21 cm and height 42 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

OR

The daily electricity consumption (in units) of households in a colony is given below. 3

Units consumed	80–90	90–100	100–110	110–120	120–130	130–140
Number of households	3	6	13	15	7	6

Find the median.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 7, 14, x, 20, y, 19, 3 respectively, and total frequency 100, is 54. Find the value of x. 5

OR

A box contains 40 cards numbered 1 to 40. One card is drawn at random. Find the probability that the number is: 5
(i) a multiple of 3,
(ii) a prime number,
(iii) a multiple of 4 (in this order).

27. A conical vessel of radius 6 cm and height 16 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. 5

OR

The mean of a distribution with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 and frequencies 3, 6, p, 15, 7, 6 respectively is 112. Find the value of p. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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